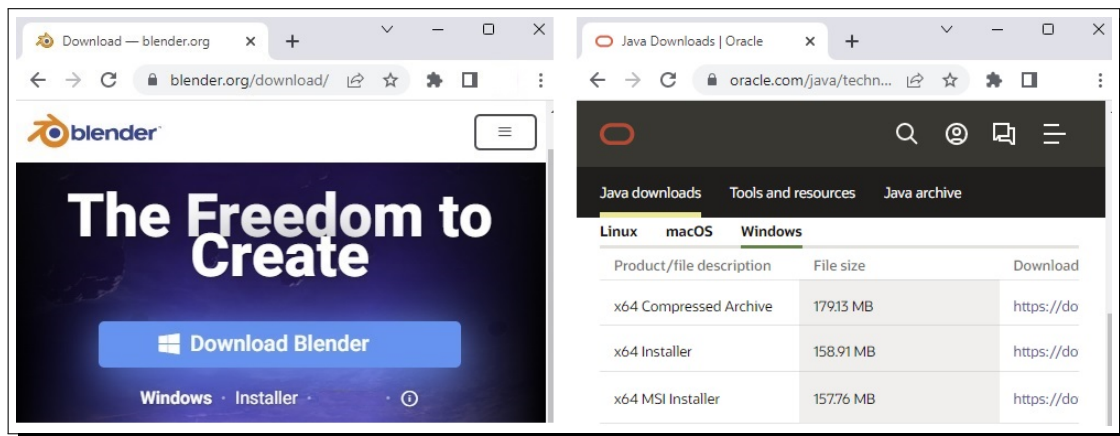


SOFTWARE INSTALLATION ON WINDOWS

1. Blender Simulation Software

- Go to [Blender Website](https://blender.org) and download the latest version of installer.
- Run the executable installer and follow the instructions for installation.

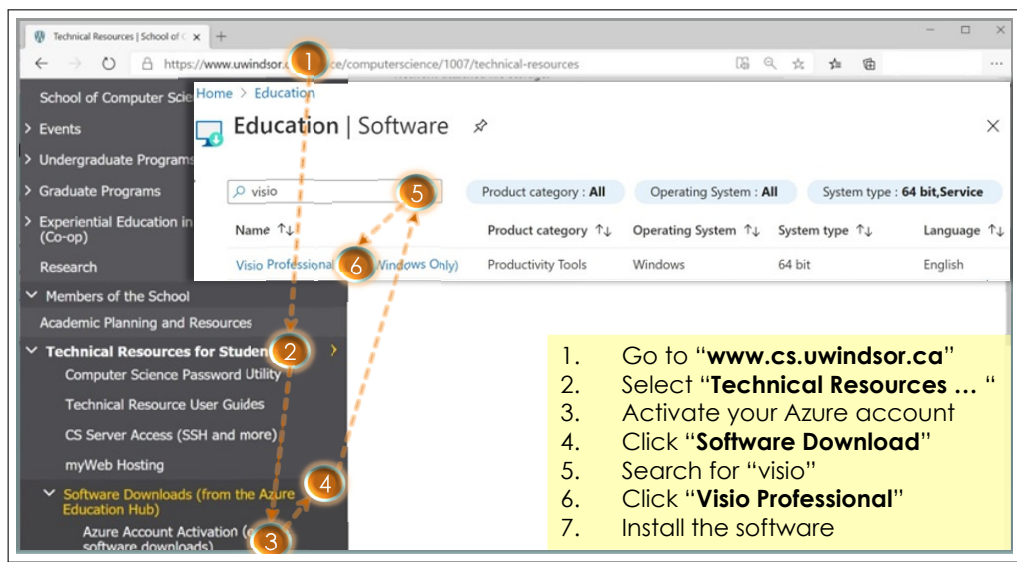


2. Java Development Kit (JDK)

- (a) Go to [Oracle Website](https://www.oracle.com/technetwork/java/javase/downloads) and download the latest version of JDK installer.
- (b) Run the executable installer and follow the instructions for installation.

3. (optional) Visio Professional

- Connect to the University network via GlobalProtect (refer to [this link](#) for installation)
- Follow the instructions in the figure below to download and install the latest version of Visio Pro available at the Azure Ecuation Hub.

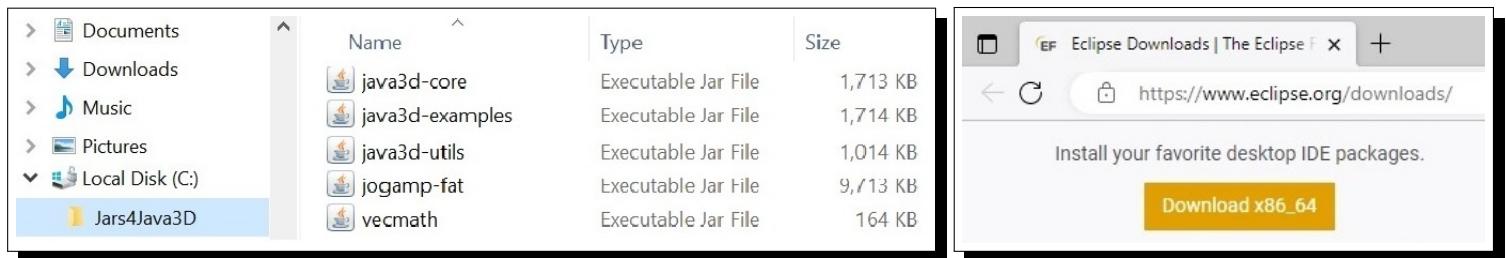


Index of /deployment/java3d/1.7.1-final

Name	Last modified	Size	Description
Parent Directory			
java3d-core-javadoc.jar	2023-09-06 00:26	2.0M	
jogamp-java3d-1.7.1-final.zip	2023-09-06 06:16	4.3M	

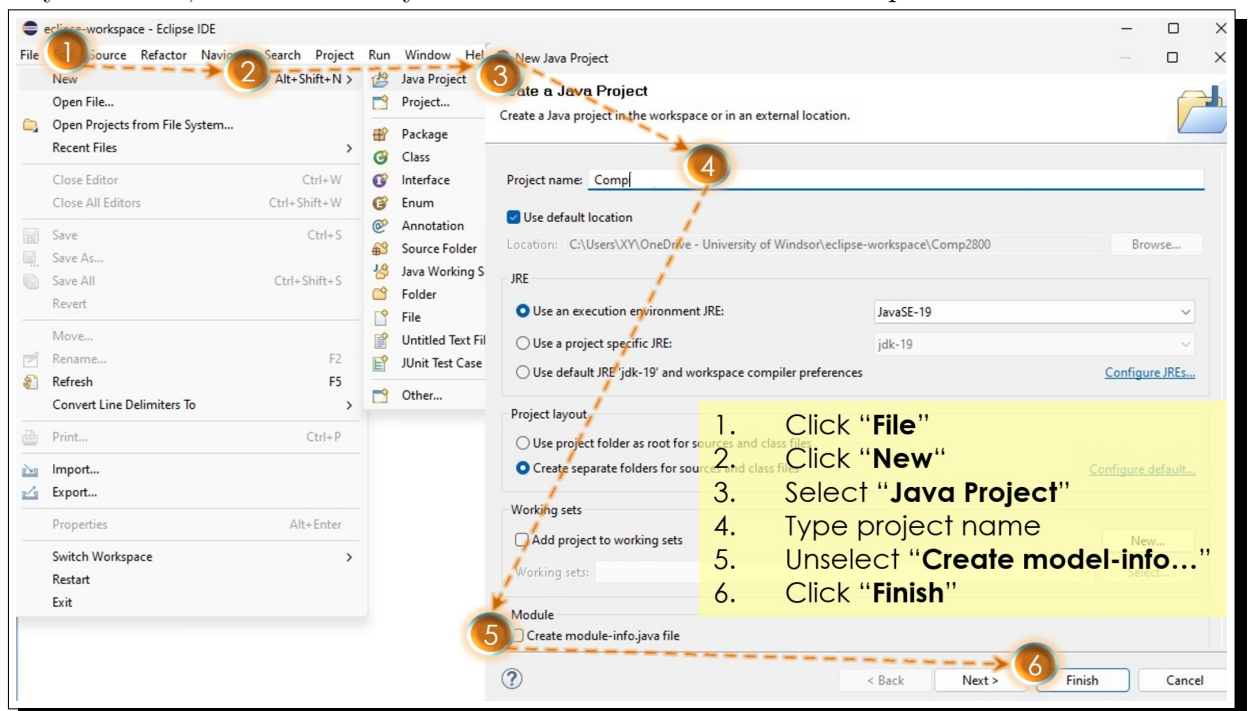
4. Java 3D Installation

- The Jars for Java3D are retrievable in a zipped file (shown in the above figure) available at [this link](#) for the latest stable version. The JogAmp fat JAR is also retrievable [here](#) as jogamp-fat-all.7z file. The jar files in both zip files can be extracted with 7-Zip.
- Create a folder under the C:/ and name it Jars4Java3D, i.e., C:/Jars4Java3D.
- Extract to this folder all the jar files in jogamp-java3d1.7.1-final.7z, including j3d-core.jar, j3d-examples.jar, j3d-utils.jar, and vecmath.jar.
- Extract to the same folder the jogamp-fat.jar file in jogamp-fat-all.7z.



5. Eclipse Installation and Setup

- Download the installer from www.eclipse.org/downloads/ as shown in the above figure.
- Start *eclipse* after the installer finishes installing “Eclipse IDE for Java Developers”.
- Create a new Java project as Comp8490XY, where XY needs to be replaced by the initials of your name, and set JRE by default to the one installed in Step 1.



- (d) After right clicking the newly created project (e.g., Comp8490XY), select **Build Path** and then **Configure Build Path** in the pop up menus. Afterwards, follow the steps to add three VM arguments, which sets `java.base/java.lang`, `java.desktop/sun.awt`, and `java.desktop/sun.java2d` to ALL-UNNAMED.

1. Right click the package name
2. Select "Build Path"
3. Select "Configure Build Path..."
4. Expand "JRE System Library"
5. Select "Is modular"

6. Click "Edit"
7. Click "Details"
8. Click "Add"
9. Fill in and add three pairs of values
10. Finish with two clicks 'a' and 'b'

- (e) While still in the pop-up window of **Configure Build Path**, select **Classpath** under the **Libraries** tab in the window opened up by clicking **Java Build Path**.

- (f) Choose to **Add External JARs** and open the jar files after locating them in the folder. The setup is completed after applying the changes and closing the popup windows.
- (g) Create a java class file and name it to **RotatingCube.java**, whose codes can be copied from the file posted in **Resources->Java3D Files** folder.
- (h) Installation is successful if **RotatingCube.java** program renders a spinning colored cube as shown in the right figure. Pressing the arrow keys should change the cube's location/size. Pressing the '=' key brings the cube back to its original location.
- (i) Online documentation for the Java3D classes is available by following this [link](#).

